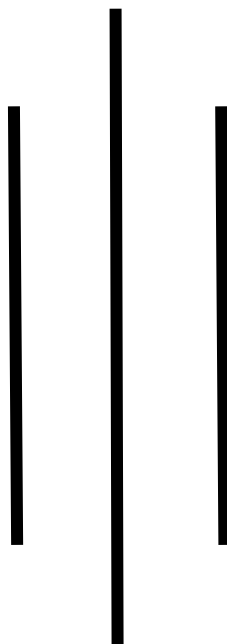




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LAB REPORT II

Lab Report on
**“TO DEMONSTRATE THE USE OF
EVENT HANDLING IN JAVA”**

Submitted By:

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Roll No: **42**

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Name: **Er. Shiva Ram Dam**

Subject: **Computer Programming II**

Subject Code: **CIT 122**

Objectives:

The main objectives of this practical is to understand and implement the usage of events and event handling using OOP concepts.

Tools and Technologies Used:

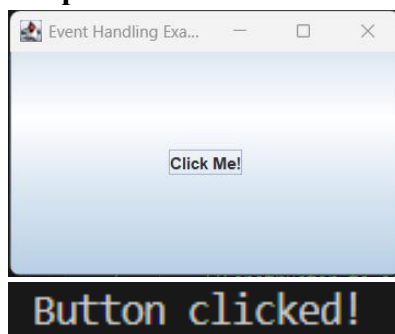
The following tools and technologies were used during this practical:

- **Operating System:** Windows 11
- **Code Editor / IDE :** Visual Studio Code
- **Language:** Java
- **Runtime:** JRE v21.0.2

1. Write a java program with a button and display “Button Clicked” when the button is clicked using Event Listener.

```
Event.java
1  import javax.swing.*;
2  import java.awt.*;
3  import java.awt.event.*;
4
5  public class Event {
6      //Constructor to set up the GUI components
7      public Event() {
8          JFrame frame = new JFrame("Event Handling Example");
9          frame.setSize(300,200);
10         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
11
12         //Create a button and add it to frame
13         JButton button = new JButton("Click Me!");
14         frame.add(button);
15         //Make the frame visible
16         frame.setVisible(true);
17         //Implement and register the event listener
18         button.addActionListener(new ActionListener() {
19             public void actionPerformed(ActionEvent e) {
20                 System.out.println("Button clicked!");
21             }
22         });
23     }
24
25     public static void main(String[] args) {
26         Event obj = new Event();
27     }
28 }
```

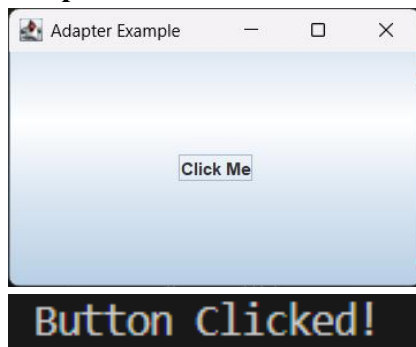
Output:



2. Write a java program with a button and display “Button Clicked” when the button is clicked using Adapter classes.

```
Event2.java
1  import java.awt.event.*;
2  import javax.swing.JButton;
3  import javax.swing.JFrame;
4
5  public class Event2 extends MouseAdapter{
6
7      public static void main(String[] args) {
8          JFrame frame = new JFrame("Adapter Example");
9          frame.setSize(300, 200);
10         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
11         JButton button = new JButton("Click Me");
12         button.addMouseListener(new MouseAdapter() {
13             public void mouseClicked(MouseEvent e) {
14                 System.out.println("Button Clicked!");
15             }
16         });
17         frame.add(button);
18         frame.setVisible(true);
19     }
}
```

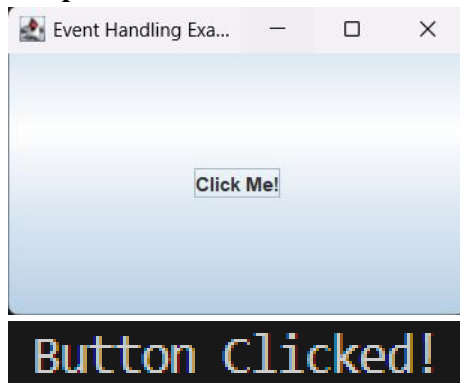
Output:



3. Write Programs with a button and display “Button Clicked” when a button is clicked using Anonymous Inner class.

```
anonymous.java
1  import javax.swing.*;
2  import java.awt.event.*;
3
4  public class anonymous {
5
6      public anonymous() {
7          JFrame frame = new JFrame("Event Handling Example");
8          frame.setSize(300, 200);
9          frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
10         JButton button = new JButton("Click Me!");
11         frame.add(button);
12         frame.setVisible(true);
13         button.addActionListener(new ActionListener(){
14             public void actionPerformed(ActionEvent e) {
15                 System.out.println("Button Clicked!");
16             }
17         });
18     }
19     public static void main(String[] args) {
20
21         anonymous obj = new anonymous();
22     }
23 }
```

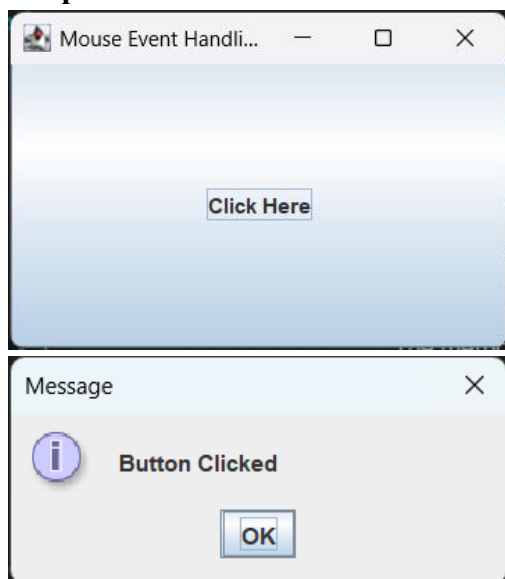
Output:



4. WAP to create a button “Click Here” in a frame which when clicked displays “Button Clicked in a message dialogue box.

```
Event4.java
1  import javax.swing.*;
2  import java.awt.event.*;
3  import java.awt.*;
4
5  public class Event4 {
6      private JButton btnClick;
7      Event4() {
8          JFrame frame = new JFrame("Mouse Event Handling");
9          frame.setSize(300, 200);
10         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
11
12         btnClick = new JButton("Click Here");
13         frame.add(btnClick);
14         frame.setVisible(true);
15
16         btnClick.addActionListener(new ActionListener() {
17             public void actionPerformed(ActionEvent arg0) {
18                 JOptionPane.showMessageDialog(null, "Button Clicked");
19             }
20         });
21     }
22     public static void main(String[] args) {
23         Event4 obj = new Event4();
24     }
25 }
```

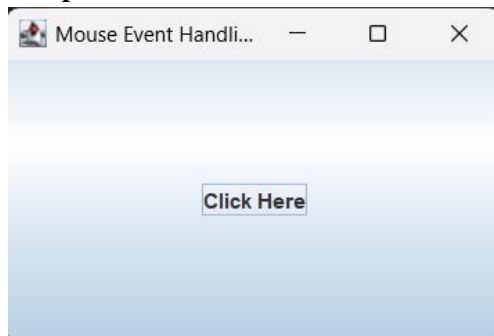
Output:



5. WAP to create a button “Click Here” in a frame which when clicked quits the application.

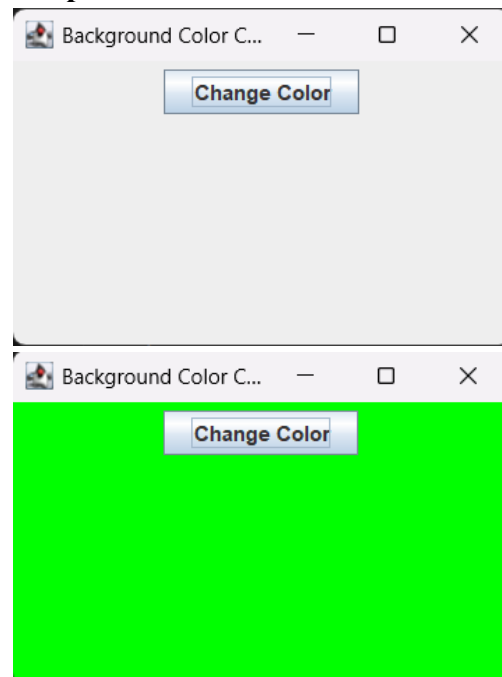
```
quitEvent.java
1  import javax.swing.*;
2  import java.awt.event.*;
3  import java.awt.*;
4
5  public class quitEvent {
6      private JButton btnClick;
7      quitEvent() {
8          JFrame frame = new JFrame("Mouse Event Handling");
9          frame.setSize(300, 200);
10         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
11
12         btnClick = new JButton("Click Here");
13         frame.add(btnClick);
14         frame.setVisible(true);
15
16         btnClick.addActionListener(new ActionListener() {
17             public void actionPerformed(ActionEvent arg0) {
18                 System.exit(0);
19             }
20         });
21     }
22     public static void main(String[] args) {
23         quitEvent obj = new quitEvent();
24     }
25 }
```

Output:



6. WAP with a button; change the background color to green on clicking the button.

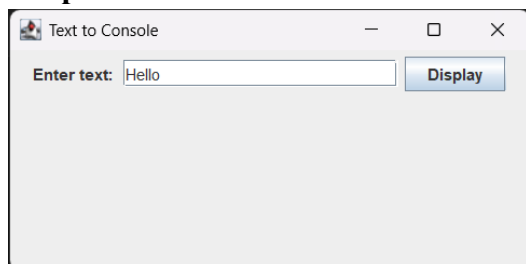
```
bgChangeEvent.java
1  import javax.swing.*;
2  import java.awt.event.*;
3  import java.awt.*;
4
5  public class bgChangeEvent {
6      private JButton btnClick;
7      bgChangeEvent() {
8          JFrame frame = new JFrame("Background Color Change");
9          frame.setSize(300, 200);
10         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
11
12         final Container cont = frame.getContentPane();
13         cont.setLayout(new FlowLayout());
14
15         btnClick = new JButton("Change Color");
16         cont.add(btnClick);
17
18         btnClick.addActionListener(new ActionListener() {
19             public void actionPerformed(ActionEvent arg0) {
20                 cont.setBackground(Color.GREEN);
21             }
22         });
23         frame.setVisible(true);
24     }
25     public static void main(String[] args) {
26         bgChangeEvent obj = new bgChangeEvent();
27     }
28 }
```

Output:

7. WAP with a textField and button. When the button is clicked, the input string is displayed.

```
textFieldEvent.java
1  import javax.swing.*;
2  import java.awt.*;
3  import java.awt.event.ActionEvent;
4  import java.awt.event.ActionListener;
5
6  public class textFieldEvent {
7      JTextField txtString;
8      JButton btnClick;
9
10     public textFieldEvent() {
11         JFrame frame = new JFrame("Text to Console");
12         frame.setSize(400, 200);
13         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
14         Container container = frame.getContentPane();
15         container.setLayout(new FlowLayout());
16         txtString = new JTextField(20);
17         btnClick = new JButton("Display");
18         container.add(new JLabel("Enter text: "));
19         container.add(txtString);
20         container.add(btnClick);
21         btnClick.addActionListener(new ActionListener() {
22             public void actionPerformed(ActionEvent e) {
23                 String text = txtString.getText();
24                 System.out.println("Entered Text: " + text);
25             }
26         });
27         frame.setVisible(true);
28     }
29     public static void main(String[] args) {
30         textFieldEvent obj = new textFieldEvent();
31     }
32 }
```

Output:



Entered Text: Hello

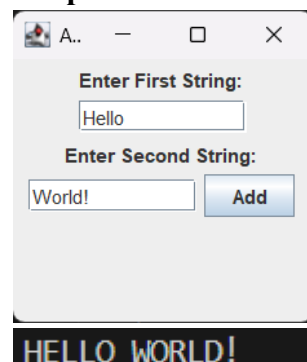
8. WAP to create two textboxes and a button. The program should concatenate input strings and display in uppercase.

```

concatenateEvent.java
1  import javax.swing.*;
2  import java.awt.event.*;
3  import java.awt.*;
4
5  public class concatenateEvent extends JFrame {
6      private JFrame frame;
7      private JTextField txtFirstString, txtSecondString;
8      private JLabel lblFirstString, lblSecondString;
9      private JButton btnChange;
10     private String result;
11
12     concatenateEvent() {
13         frame = new JFrame("Adder");
14         frame.setSize(200, 200);
15         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
16
17         Container container = frame.getContentPane();
18         container.setLayout(new FlowLayout());
19
20         lblFirstString = new JLabel("Enter First String: ");
21         lblFirstString.setBounds(50, 50, 100, 20);
22         container.add(lblFirstString);
23
24         txtFirstString = new JTextField(10);
25         txtFirstString.setBounds(200, 50, 150, 30);
26         container.add(txtFirstString);
27
28         lblSecondString = new JLabel("Enter Second String: ");
29         lblSecondString.setBounds(50, 50, 100, 20);
30         container.add(lblSecondString);
31
32         txtSecondString = new JTextField(10);
33         txtSecondString.setBounds(200, 50, 150, 30);
34         container.add(txtSecondString);
35
36         btnChange = new JButton("Add");
37         container.add(btnChange);
38
39         btnChange.addActionListener(new ActionListener() {
40             public void actionPerformed(ActionEvent e) {
41                 String st1 = txtFirstString.getText();
42                 String st2 = txtSecondString.getText();
43                 result = st1 + st2;
44                 System.out.println(result.toUpperCase());
45             }
46         });
47         frame.setVisible(true);
48     }
49     public static void main(String[] args) {
50         concatenateEvent obj = new concatenateEvent();
51     }
52 }

```

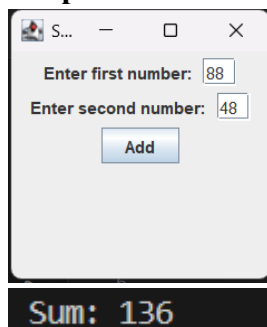
Output:



9. WAP to create a window to accept two numbers and display the sum of them when ADD button is clicked.

```
sumEvent.java
1  import javax.swing.*;
2  import java.awt.event.*;
3  import java.awt.*;
4
5  public class sumEvent extends JFrame {
6      private JFrame frame;
7      private JTextField intFirstNum, intSecondNum;
8      private JLabel lblFirstNum, lblSecondNum;
9      private JButton btnClick;
10     private int result;
11
12     sumEvent() {
13         frame = new JFrame("Sum of Two Numbers");
14         frame.setSize(200, 200);
15         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
16
17         Container container = frame.getContentPane();
18         container.setLayout(new FlowLayout());
19
20         lblFirstNum = new JLabel("Enter first number: ");
21         container.add(lblFirstNum);
22
23         intFirstNum = new JTextField(2);
24         container.add(intFirstNum);
25
26         lblSecondNum = new JLabel("Enter second number: ");
27         container.add(lblSecondNum);
28
29         intSecondNum = new JTextField(2);
30         container.add(intSecondNum);
31
32         btnClick = new JButton("Add");
33         container.add(btnClick);
34
35         btnClick.addActionListener(new ActionListener() {
36             public void actionPerformed(ActionEvent e) {
37                 int num1 = Integer.parseInt(intFirstNum.getText());
38                 int num2 = Integer.parseInt(intSecondNum.getText());
39                 result = num1 + num2;
40                 System.out.println("Sum: " + result);
41             }
42         });
43         frame.setVisible(true);
44     }
45
46     public static void main(String[] args) {
47         sumEvent obj = new sumEvent();
48     }
```

Output:

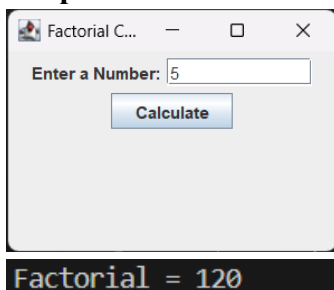


10. WAP to create a window to calculate factorial of an entered number.

```

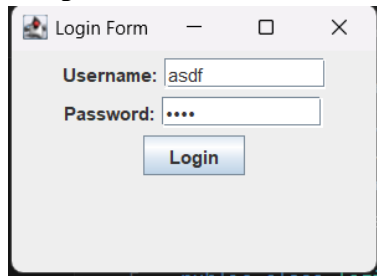
factorialEvent.java > ...
1  import javax.swing.*;
2  import java.awt.*;
3  import java.awt.event.*;
4
5  public class factorialEvent extends JFrame {
6      private JFrame frame;
7      private JLabel lblNumber;
8      private JTextField Num;
9      private JButton btnClick;
10     private long fact = 1;
11
12     factorialEvent() {
13
14         frame = new JFrame("Factorial Calculator");
15         frame.setSize(250, 180);
16         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
17
18         Container container = frame.getContentPane();
19         container.setLayout(new FlowLayout());
20
21         lblNumber = new JLabel("Enter a Number:");
22         container.add(lblNumber);
23
24         Num = new JTextField(10);
25         container.add(Num);
26
27         btnClick = new JButton("Calculate");
28         container.add(btnClick);
29
30         btnClick.addActionListener(new ActionListener() {
31
32             public void actionPerformed(ActionEvent e) {
33
34                 int num = Integer.parseInt(Num.getText());
35
36                 fact = 1;
37
38                 for(int i = 1; i <= num; i++) {
39                     fact = fact * i;
40                 }
41                 System.out.println("Factorial = " + fact);
42             }
43         });
44         frame.setVisible(true);
45     }
46     Run main | Debug main
47     public static void main(String[] args) {
48         factorialEvent obj = new factorialEvent();
49     }

```

Output:

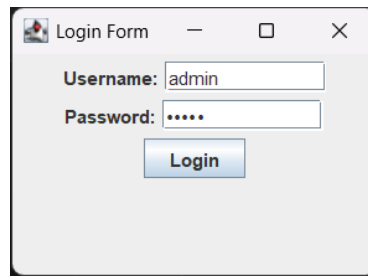
11. WAP to display “Welcome to Homepage” after enter valid username and password. Assume the valid username and password are “admin” and “12345”.

```
loginEvent.java
1  import javax.swing.*;
2  import java.awt.*;
3  import java.awt.event.*;
4
5  public class loginEvent extends JFrame {
6
7      private JFrame frame;
8      private JLabel lblUsername, lblPassword;
9      private JTextField txtUsername;
10     private JPasswordField txtPassword;
11     private JButton btnLogin;
12
13     loginEvent() {
14
15         frame = new JFrame("Login Form");
16         frame.setSize(250, 180);
17         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
18
19         Container container = frame.getContentPane();
20         container.setLayout(new FlowLayout());
21
22         lblUsername = new JLabel("Username:");
23         container.add(lblUsername);
24
25         txtUsername = new JTextField(10);
26         container.add(txtUsername);
27
28         lblPassword = new JLabel("Password:");
29         container.add(lblPassword);
30
31         txtPassword = new JPasswordField(10);
32         container.add(txtPassword);
33
34         btnLogin = new JButton("Login");
35         container.add(btnLogin);
36
37         btnLogin.addActionListener(new ActionListener() {
38
39             public void actionPerformed(ActionEvent e) {
40
41                 String username = txtUsername.getText();
42                 String password = txtPassword.getText();
43
44                 if (username.equals("admin") && password.equals("12345")) {
45                     System.out.println("Welcome to HomePage");
46                 } else {
47                     System.out.println("Invalid Username or Password");
48                 }
49             }
50         });
51
52         frame.setVisible(true);
53     }
54     public static void main(String[] args) {
55         loginEvent obj = new loginEvent();
56     }
57 }
```

Output:

A Java Swing window titled "Login Form" with standard window controls. It contains two text input fields: "Username:" with the text "asdf" and "Password:" with five dots. Below the fields is a "Login" button.

Invalid Username or Password



A Java Swing window titled "Login Form" with standard window controls. It contains two text input fields: "Username:" with the text "admin" and "Password:" with five dots. Below the fields is a "Login" button.

Welcome to HomePage